

Lecturer's name:		Date:	
Course: Science 8		Duration:	
Unit:		TEKS: TEKS 8.2 A-E	
Lesson objectives:		By the end of the lesson, students will be able to plan, conduct, and analyze a scientific experiment Students will use this knowledge to plan and carry out an experiment.	
Essential Questions:		What techniques and methods go into the scientific method? How does one synthesize scientific data to draw conclusions?	
LYS Methodology Be (Character/Values/Principles) - Know (What resources are available to the student/ How can they use them) - Do (Execute with Excellence) -		Be: Why is it important to be understanding? Know: What is self esteem? Do: What steps can you take to develop your self esteem?	
Resources(Include Technology):		Lesson Wide Laptop Projector Instruction The scientific method (video) Khan Academy https://www.khanacademy.org/science/high-school-biology/hs-biology-foundations/hs-biology-and-the-scientific-method/v/the-scientific-method Activity Bacteria Growth Virtual Lab https://biomanbio.com/HTML5GamesandLabs/SciMethodGames/bacterialabpage.html Lab Reporting Organizer	

	https://worksheetplace.com/index.php?function=DisplaySheet&sheet=Lab-reporting-organizer&links=2&id=16805&link1=241&link2=451
Instructional Input - Asynchronous AS: Anticipatory Set (Introduction) M: Modeling (I do) GP: Guided Practice (We do)	As: Students will write a few sentences, up to a paragraph, on what they think the Scientific Method is. Students will upload their answers to the LMS. M: Students will watch a video on the Scientific Method. GP: Students will carry out the Bacteria Growth Virtual Lab. Before and during the experimental students will fill out the Lab Reporting Organizer.
Instructional Input - Synchronous AS: Anticipatory Set (Introduction) M: Modeling (I do) GP: Guided Practice (We do)	As: Students will write a few sentences, up to a paragraph, on what they think the Scientific Method is. Students will upload their answers to the LMS. M: Instructor will lecture on the Scientific Method. GP: Students will carry out the Bacteria Growth Virtual Lab. Before and during the experimental students will fill out the Lab Reporting Organizer.
Independent Practice/ Homework	Students will complete a one page write up discussing how they planned the experiment and then carried out the experiment. Students will also write a short analysis of the experiment.
Lesson Close What is the answer to the essential question, and why does it matter to me in terms of: Educational? Social? Cultural? Financial? Health? Vocational? Spiritual?	Spiritual Just as scientists use the scientific method to understand the world around them, you too need to take steps to understand the world around you and improve yourself as a person. You need to be able to understand what motivates others. People might have different motivations or goals than you, and that's okay, but you need to be understanding, especially if someone comes from a different background than you. Why is it important to be understanding? Social Just as scientists believe in their knowledge and abilities, you too need to believe in your capabilities and have strong self esteem. What are some things that help your self esteem? Social Everyone has doubts, everyone at some point questions their abilities. Despite these doubts, it is important to believe in yourself. Having a bad day is normal, but that doesn't make you any lesser of a person. What are some activities you can participate in to boost your self esteem?